The latest spectroscopy products for 2026

Spectroscopy is the study of the absorption and emission of light by matter (Image credit: Forance/Shutterstock.com)

Spectroscopy is a powerful tool in science and technology. It is applied across various fields, including chemical analysis, astronomy, medicine and biology, structural biology, environmental science, material science, forensics, pharmaceuticals, and industrial applications, to name a few. In all these applications, spectroscopy can provide insights into the physical, chemical, and structural properties of matter, often with high precision and minimal sample preparation.

What is spectroscopy?

Spectroscopy is the study of the absorption and emission of light by matter. It involves splitting the light into its constituent wavelengths in a similar way to a prism.

The technique is most often used to study the structures of atoms and molecules, which emit large numbers of wavelengths and so allow for the detailed investigation of their structures. It is widely used in science and academia thanks to benefits such as

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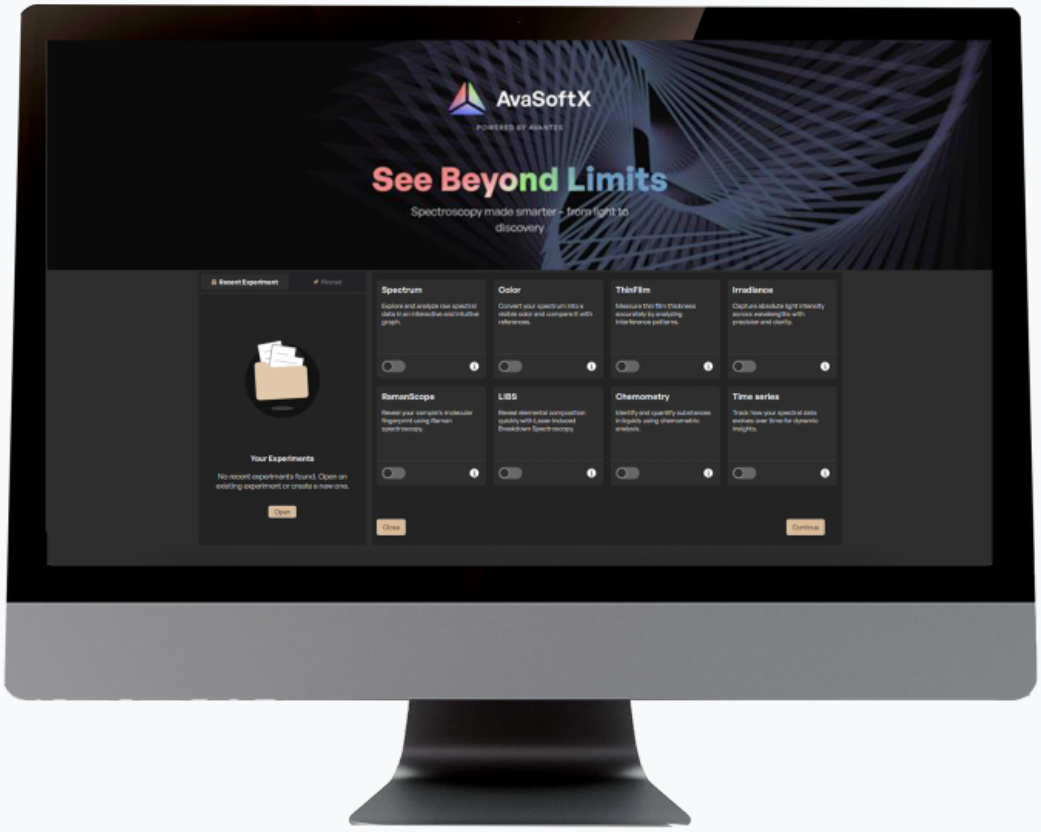
sensitivity, safety, non-invasiveness, cost-effectiveness and sample speed. It is a broad field that has a number of sub-disciplines, each with different spectroscopic techniques.



AvaSoftX – Smarter spectroscopy starts here

AvaSoftX is Avantes' powerful new spectroscopy software platform, designed to streamline the journey from raw spectra to actionable results. Tailored for researchers and integrators alike, AvaSoftX features intuitive mini-apps for key techniques such as LIBS, Raman, and chemometry.

Step-by-step smart wizards guide users through measurement setup, from instrument configuration to referencing and data acquisition. For advanced users, AvaSoftX enables full customization through scripting and user-defined libraries, supporting complex or repeatable analysis routines.

Designed for both lab and industrial use, AvaSoftX features a clean, dark-themed interface optimized for long sessions. Embedded instructional videos and manuals ensure guidance is always close at hand.

Visit the Avantes website to explore how AvaSoftX brings clarity, control, and confidence to your spectroscopy workflow.

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What are the types of spectroscopy?

Types of spectroscopy include optical spectroscopy, which uses visible, UV, or IR light to study substances; mass spectroscopy, which measures the mass-to-charge ratio of ions to analyse chemical composition; nuclear magnetic resonance (NMR), which examines atomic nuclei in magnetic fields to reveal molecular structures; and X-ray Spectroscopy, which studies the interaction of X-rays with matter to analyse atomic structures.

Spectroscopy: the advantages and disadvantages

As stated, spectroscopy brings many advantages to science and academia, but it does also have its limitations.

From sensors to spectrometers: A unique in-house expertise in spectral analysis

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Hamamatsu Photonics offers a versatile range of spectroscopy products for precise light analysis across a wide variety of applications. The portfolio includes mini-spectrometers for portable systems used in environmental monitoring, quality control, and process analysis, including the OPAL Luxe® series, known for its excellent dynamic range and sensitivity.

The newest addition is the WS Series mini-spectrometers, designed to combine compact size, high spectral resolution, and broad UV to NIR coverage in a lightweight enclosure. By integrating a reflective grating optical system, a high-sensitivity CMOS sensor, and driver circuitry, the WS Series is well suited for modern spectroscopy applications in both research and industry.

For mid-infrared spectroscopy, Hamamatsu's MEMS FPI and MEMS FTIR modules provide intelligent solutions for tunable wavelength detection and precise molecular analysis, enabling gas sensing, biomedical diagnostics, and material characterisation. High-quality
image sensors, developed and manufactured in-house, ensure superior spectrometer performance with outstanding sensitivity and speed.

Together, these technologies enable compact, reliable, and scalable designs tailored to diverse industry needs.

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Advantages

Qualitative analysis: it can identify unknown substances.
Quantitative analysis: it can determine the concentration of specific compounds.
Structural determination: It provides insights into molecular structure.
Non-destructive analysis: It often doesn't damage the sample.
Versatility: It is adaptable to various fields and applications.
Remote sensing: It can analyse substances from a distance.
Medical diagnostics: It is used in medical imaging and diagnostics.

Disadvantages

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Cost: equipment for spectroscopy can be expensive.
Complexity: some techniques can require specialist training and expertise.
Sample preparation: Some methods require sample preparation, which can be time-consuming.
Interference: certain substances can interfere with the analysis.

What optical components are used for spectroscopy?

Optical components for spectroscopy include light sources, such as halogen, xenon and deuterium, depending on the wavelength range and application. Non-dispersive materials are used to filter out any non-targeted ranges of wavelengths from the source to improve resolution, while dispersive materials used include diffraction gratings and prisms. Other components include optical fibre cables.

Spectroscopy products on the market now

Vendors of products and solutions for spectroscopy include:

Armadillo SIA is a leading manufacturer of specialised optical fibre and hybrid photonic solutions. Its fibres are critical in chemical processing and pharmaceutical testing, with its NIR spectroscopy bundles seeing high demand in food analysis.

Art Photonics specialises in the development and production of speciality fibre products from 300nm to 16µm. Its polycrystalline mid-infrared (PIR-) fibres are used for the assembly of various spectroscopy probes for medical diagnostics and industrial process control.

Avantes specialises in high-end fibre-optic spectroscopy systems. Its CompactLine and SensLine ranges are core offerings for measurements in the ultra-violet, visible, and near-infrared wavelengths, while its NIRLine and StarLine spectrometers are designed for versatile general applications.

CeNing Optics is an optical company that produces high-quality components, including prisms, lenses, and filters. It continues to support both standard and custom designs for spectroscopy systems.

Coherent offers a broad range of spectroscopy lasers designed for LIBS, Raman, and MALDI-TOF. Its flare ytterbium laser is a top choice for maximising TOF resolution, while its femtosecond laser amplifiers lead in solid sample analysis.

Delta Photonics is a distributor of lasers, optics, and scientific cameras. It provides spectroscopy equipment for applications such as fluorescence and plasma emission, carrying products from partners including Avantes, Edinburgh Instruments, and Teledyne Princeton Instruments.

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GWU-Lasertechnik offers a range of CCD-based spectrometers covering the UV to mid-infrared (2μm). Its thermo-electric (TE) cooled devices are particularly noted for ultra-low noise in weak-light applications like fluorescence sensing.

Hamamatsu Photonics offers more than 20 different types of spectrometers. Its palm-sized FTIR engine is a standout in 2026, as it assists in the creation of portable, handheld FTIR spectrophotometers for analytical applications.

Horiba Scientific offers high-quality spectrometers, monochromators, and spectrographs. Its spectroscopy products span Raman, fluorescence, and XRF, and it also provides critical components such as diffraction gratings.

Ibsen Photonics is a partner for high-volume OEM supply of spectrometers and transmission gratings. Its Rock, Freedom, Pebble, Peak, and Eagle series are industry-standard configurations for robust, industrial-grade OEM spectrometers.

NKT Photonics offers broadband white light lasers. Its SuperK Fianium supercontinuum series is as broadband as a lamp and as bright as a laser, ensuring a maintenance-free lifetime of thousands of hours.

Manx Precision Optics is a manufacturer of precision optical components, including mirrors and etalons. It maintains a large stock of raw material and polished optics for ultrafast applications.

Ocean Optics offers compact spectrometers covering UV, visible, and NIR wavelengths. It has focused on integrating the latest advances in opto-electronic design into its newest instrument range.

OtO Photonics has a wide range of spectrometers, including the ultra-compact Bullet series. Its GoldenBullet NIR model is now a staple of its lineup, offering two-stage TE-cooling for ranges up to 2200nm.

Photonic Solutions provides a suite of modular spectroscopy equipment and photovoltaic testing systems. As the exclusive UK distributor for Becker & Hickl, it offers a complete portfolio of time-correlated single photon counting (TCSPC) equipment.

Teledyne Princeton Instruments (TPI) designs and manufactures high-performance cameras and spectrographs. It continues to innovate for the scientific research community, having designed CCD cameras with exceptionally high sensitivity and low read noise.

Pro-Lite Technology carries instruments for performing optical spectroscopy, including spectral imagers and IR cameras. It also supplies specialised instruments for plant

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science and fruit ripeness measurements.

Sacher Lasertechnik is an expert in tunable high-power external cavity diode lasers. It offers various equipment for laser spectroscopy, including reference gas and vapour cells with integrated heaters.

Toptica offers industrial and research-grade lasers for spectroscopy with ultra-broad wavelength coverage. Its difference frequency combs and terahertz systems complement its portfolio for the most demanding applications.

Wasatch Photonics (a Foss company) designs spectroscopy products using its patented VPH gratings. It has worked with applications ranging from material identification to biomedical analysis, aiming to bridge the performance gap between analytical and portable devices.

This is not an exhaustive list. If you provide products and solutions in the spectroscopy space and would like your company to be included, please let us know at: editor.electro@europascience.com.

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